

1. What is Data Science? Explain its primary objective.

Data Science is an interdisciplinary field that combines statistics, mathematics, programming, machine learning, and domain knowledge to extract meaningful insights from data. It helps organizations understand patterns, predict future outcomes, and make informed decisions.

Primary Objective:

- To collect, process, analyze, and interpret data.
- To discover hidden patterns and trends.
- To support data-driven decision-making.
- To solve real-world problems using data.

2. Why is Data Science important in today's digital world? Mention any four reasons.

Data Science has become essential because huge amounts of data are generated every day through social media, online shopping, banking, healthcare, and IoT devices.

Importance:

- Helps organizations make better business decisions.
- Predicts future trends using historical data.
- Improves customer experience through personalization.
- Automates business processes using AI and Machine Learning.
- Detects fraud and enhances security.

3. List any four benefits of Data Science in business and society.

Data Science offers several advantages by converting raw data into useful information.

Benefits include:

- Better and faster decision-making.

- Improved customer satisfaction through personalized services.
- Fraud detection and risk management in banking.
- Enhanced healthcare through disease prediction and diagnosis.
- Increased business efficiency and productivity.

4. Mention four real-world applications of Data Science.

Data Science is widely used in various industries.

Some important applications are:

- **Healthcare:** Disease prediction and medical diagnosis.
- **Banking:** Credit scoring and fraud detection.
- **E-commerce:** Product recommendation systems.
- **Transportation:** Traffic prediction and route optimization.
- **Agriculture:** Crop yield prediction and smart farming.

5. What are the different facets of data? Briefly explain each.

Data can be classified into different forms depending on its structure.

- **Structured Data:** Organized in rows and columns with a fixed format. Example: Employee database.
- **Semi-Structured Data:** Does not follow a strict table format but contains tags or labels. Example: XML and JSON files.
- **Unstructured Data:** Has no predefined structure. Example: Images, videos, emails, social media posts.

Understanding these facets helps in selecting suitable tools for data analysis.

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Structured Data	Semi-Structured Data	Unstructured Data
Stored in tables	Stored with tags	No fixed format
Easy to search	Moderately organized	Difficult to process
Uses relational databases	Uses XML/JSON	Includes images, videos, text
Example: Student database	Example: XML file	Example: YouTube videos

7. What is Big Data? State its key characteristics (5 Vs).

Big Data refers to extremely large and complex datasets that cannot be processed efficiently using traditional database systems.

The **5 Vs of Big Data** are:

- **Volume:** Huge amount of data generated.
- **Velocity:** Speed at which data is generated and processed.
- **Variety:** Different types of data such as text, images, and videos.
- **Veracity:** Accuracy and reliability of data.
- **Value:** Useful insights obtained from data.

8. Explain the Big Data ecosystem. Name any four technologies used in it.

The **Big Data ecosystem** consists of technologies that help collect, store, process, analyze, and visualize large datasets. It enables organizations to manage massive amounts of structured and unstructured data efficiently.

Technologies used include:

- Hadoop
- Apache Spark

- Hive
- HBase
- Kafka
- MongoDB

9. What are the major phases of the Data Science process?

The Data Science process generally consists of the following phases:

1. Data Collection
2. Data Cleaning and Preparation
3. Data Exploration and Analysis
4. Feature Engineering
5. Model Building
6. Model Evaluation
7. Model Deployment
8. Monitoring and Maintenance

Each phase contributes to developing accurate and reliable data-driven solutions.

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of collecting or accessing data from different sources for analysis and decision-making.

Common data sources include:

- Relational databases (MySQL, PostgreSQL)
- APIs (Application Programming Interfaces)
- Data warehouses

- CSV and Excel files
- Websites through web scraping

11. Why is data collection considered an important step in the Data Science process?

Data collection is the foundation of every Data Science project. The quality of collected data directly affects the performance of the final model.

Its importance includes:

- Provides accurate and relevant information.
- Improves model accuracy.
- Reduces errors during analysis.
- Supports reliable decision-making.

12. What is data cleansing? List any four common data-cleaning techniques.

Data cleansing is the process of identifying and correcting inaccurate, incomplete, or inconsistent data before analysis.

Common techniques include:

- Removing duplicate records.
- Handling missing values.
- Correcting inconsistent formats.
- Removing outliers.
- Standardizing data values.

Data cleansing improves the quality and reliability of analysis.

13. What is data integration? Why is it necessary in Data Science?

Data integration is the process of combining data from multiple sources into a single, consistent dataset.

It is necessary because it:

- Eliminates duplicate information.
- Improves data consistency.
- Provides a complete view of the data.
- Enables accurate analysis and reporting.
- Supports better business decisions.

14. Explain data transformation with suitable examples.

Data transformation is the process of converting raw data into a suitable format for analysis or machine learning.

Examples include:

- Converting text values into numerical values.
- Scaling data between 0 and 1.
- Changing date formats.
- Aggregating monthly sales into yearly sales.

Proper transformation improves model performance and analysis accuracy.

15. What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining, organizing, and interpreting data to discover useful information and patterns.

Common techniques include:

- Descriptive Analysis
- Statistical Analysis

- Regression Analysis
- Cluster Analysis
- Data Visualization

These techniques help organizations make informed decisions.

16. Differentiate between descriptive, predictive, and prescriptive analytics.

Descriptive Analytics	Predictive Analytics	Prescriptive Analytics
Describes past events	Predicts future outcomes	Recommends best actions
Uses historical data	Uses machine learning and statistics	Uses optimization and AI
Answers "What happened?"	Answers "What will happen?"	Answers "What should be done?"
Example: Sales reports	Example: Demand forecasting	Example: Inventory optimization

17. What is meant by building a model in Data Science? Mention the basic steps involved.

Model building is the process of creating a machine learning model that learns patterns from historical data to make predictions or classifications.

Basic steps include:

1. Collect and prepare data.
2. Select suitable features.
3. Choose an appropriate algorithm.
4. Train the model.
5. Test and evaluate the model.

6. Fine-tune the model for better performance.

18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

Model evaluation measures how well a machine learning model performs on unseen data. It helps compare different models and ensures that predictions are accurate and reliable.

Common evaluation metrics are:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC (optional)

Proper evaluation helps select the best-performing model.

19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

Presenting findings helps stakeholders understand the analysis and make informed decisions. Visual representations make complex data easier to interpret.

Common visualization tools and techniques include:

- Tableau
- Microsoft Power BI
- Matplotlib
- Microsoft Excel Charts
- Bar Charts
- Line Graphs

- Pie Charts
- Scatter Plots

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

After successful testing, Data Science models are deployed using web applications, mobile apps, cloud platforms, or APIs so that users can access predictions in real time.

Applications include:

- Recommendation systems (Netflix, Amazon)
- Fraud detection in banking
- Spam email filtering
- Medical diagnosis systems
- Weather forecasting
- Self-driving vehicles

Deployment ensures that the model can continuously provide predictions and support real-world decision-making